MAIS PAC

IPhO 02.13: Quantum Physics: Waves, Atoms and Spectra · Worked Solutions

Quantum Physics: Waves, Atoms and Spectra: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Solution 1 — A photon's energy. Return →

(a) E = hc/λ = (6.63×10^(-34))(3.00×10^8)/(550×10^(-9)) = 3.62×10^(-19) {J} = 2.26 {eV}; p = E/c = 1.21×10^(-27) {kg,m,s^(-1)}.

(b) E = 1240/200 = 6.20 {eV} = 9.93×10^(-19) {J}; p = 3.31×10^(-27) {kg,m,s^(-1)}.

(c) E = 1240/0.10 = 1.24×10^4 {eV} = 12.4 {keV}; p = 6.62×10^(-24) {kg,m,s^(-1)}. **Check:** using hc = 1240 {eV,nm} gives every answer in one step, and p = E/c is consistent with p = h/λ.

Solution 2 — The photoelectric effect. Return →

(a) E = 1240/400 = 3.10 {eV}.

(b) K_(max) = 3.10 - 2.30 = 0.80 {eV}, so the stopping voltage is 0.80 {V}.

(c) λ_0 = hc/φ = 1240/2.30 = 539 {nm}. **Check:** 400 {nm} is shorter than 539 {nm}, so emission does occur — consistent with (b).

Solution 3 — Measuring Planck's constant. Return →

(a) eV = hc/λ - φ, so subtracting the two measurements:

$$e(1.20 - 0.50) = hc((1)/(300)-(1)/(400))\times10^{9}. \quad (1)$$

The bracket is 8.33×10^{5} {m^(-1)}, so h = $((1.60\times10^{-19})(0.70))/((3.00\times10^{8})(8.33\times10^{5})) = 4.5\times10^{-34}$ {J,s} — the right order, and a real experiment with more points does better.

(b) φ = hc/λ - eV = (1240/400) {eV} - 0.50 {eV} = 3.10 - 0.50 = 2.60 {eV}.

(c) $\lambda_0 = 1240/2.60 = 477$ {nm}. The slope of K against f is h for every metal; only the intercept -φ changes, which is why the graph is a family of parallel lines. **Check:** with the approximate h found in (a), hc/e = 1350 {eV,nm}; then 1350/477 = 2.83 {eV}, close to 2.60 {eV} given the crude two-point data.

Solution 4 — De Broglie wavelength. Return →

(a) K = 100 {eV} = 1.60×10^{-17} {J}, so p = $\sqrt{(2mK)} = \sqrt{(2(9.11\times10^{-31})(1.60\times10^{-17}))} = 5.40\times10^{-24}$ {kg,m,s^(-1)} and $\lambda = h/p = 1.23\times10^{-10}$ {m} = 0.123 {nm}.

(b) At 100 {keV} the electron is relativistic ($\gamma\approx1.2$): $E_{\{total\}} = 0.611$ {MeV}, and $pc = \sqrt{(E^2-(mc^2)^2)} = \sqrt{(0.611^2-0.511^2)} = 0.335$ {MeV}, so $p = 1.79\times10^{-22}$ and $\lambda = 3.7\times10^{-12}$ {m} = 3.7 {pm}.

(c) $p = h/\lambda = 5.4\times10^{-24}$ {kg,m,s^(-1)} gives $v = p/m = 5.4\times10^{-21}$ {m,s^(-1)} for a gram — a distance smaller than a proton's radius per age of the universe. **Check:** everyday objects have λ10^(-34) {m}, which is why diffraction of a thrown ball is never observed.

Solution 5 — Hydrogen lines. Return →

(a) $(1)/(\lambda) = R((1)/(4)-(1)/(9)) = R(0.1389) = 1.52\times10^{6}$ {m^(-1)}, so $\lambda = 6.56\times10^{-7}$ {m} = 656 {nm} — the red Balmer line.

(b) The shortest Balmer wavelength is the series limit $n_i\to\infty$: $(1)/(\lambda) = R/4$, so $\lambda = 4/R = 365$ {nm}.

(c) $\Delta E = 13.6(1-(1)/(4)) = 10.2$ {eV} = 1.63×10^{-18} {J}. Momentum conservation gives the atom $p = E/c = 5.4\times10^{-27}$ {kg,m,s^(-1)}, so its recoil speed is $p/m_H = 3.2$ {m,s^(-1)}. **Check:** the recoil energy, $p^2/2m_H\approx5\times10^{-21}$ {J}, is negligible compared with the photon energy — the reason the recoil is normally ignored in atomic spectra but not in Mössbauer spectroscopy.

Solution 6 — The Bohr model. Return →

(a) Setting the Coulomb force equal to mv^2/r and using $mvr = nh/2\pi$:

$$(ke^2)/(r^2) = (mv^2)/(r),\ v = (nh)/(2\pi\, mr)\ r_n = (n^2h^2\varepsilon_0)/(\pi\, me^2) = n^2a_0, \quad (2)$$

and substituting the constants gives a_0 = 5.29×10^(-11) {m}.

(b) v_1 = (h)/(2π m_ea_0) = (6.63×10^(-34))/(2π(9.11×10^(-31))(5.29×10^(-11))) = 2.19×10^6 {m,s^(-1)} = 0.0073c — slow enough that a non-relativistic treatment is defensible, but fast enough that the electron is quite relativistic in heavier atoms.

(c) E_2 = -13.6/4 = -3.40 {eV}. **Check:** the ionisation energy from n=2 is then 3.40 {eV}, matching the Balmer series limit at 365 {nm} (1240/365 = 3.40).

Solution 7 — Compton scattering. Return →

(a) The Compton wavelength is h/(m_ec) = 2.43×10^(-12) {m}, so

$$\Delta\lambda = 2.43\times10^{(-12)}(1-\cos 90^{\circ}) = 2.43\times10^{(-12)}\ \{m\}, \quad (3)$$

giving λ' = 0.0500 + 0.0024 = 0.0524 {nm}.

(b) The photon energy falls from 1240/0.0500 = 24.8 {keV} to 1240/0.0524 = 23.7 {keV}, so the electron receives 1.1 {keV}.

(c) For visible light λ≈500 {nm}, a shift of 0.0024 {nm} is a relative change of 5×10^(-6) — below the resolution of ordinary spectrometers, and in any case the electron in a metal is not free. **Check:** the fractional shift is Δλ/λ = 2.43 {pm}/50 {pm}≈5%, large for X-rays and negligible for light — exactly the pattern that made Compton's result convincing.

Solution 8 — The uncertainty principle. Return →

(a) Δ p≈/(2Δ x) = (1.05×10^(-34))/(2×1.0×10^(-10)) = 5.3×10^(-25) {kg,m,s^(-1)}.

(b) K≈((Δ p)²)/(2m) = ((5.3×10^(-25))²)/(2(9.11×10^(-31))) = 1.5×10^(-19) {J} = 0.96 {eV}. This is smaller than 13.6 {eV}, so confinement does not destroy the atom — but it is the right order, which is why atoms have sizes of order a_0 and not smaller.

(c) For a proton, Δ p = 1.05×10^(-34)/(2×10^(-15)) = 5.3×10^(-20), so K≈((5.3×10^(-20))²)/(2(1.67×10^(-27))) = 8.4×10^(-13) {J} = 5.3 {MeV} — comparable with nuclear binding energies per nucleon, which is why nuclei are about 1 {fm} across and why confining a proton more tightly costs MeV. **Check:** using Δ xΔ p/2 with Δ x = 10^(-15) {m} reproduces the known scale of nuclear binding, a strong indication that the estimate is sound.

Solution 9 — A photon and an electron with the same energy. Return →

(a) λ = 1240/2000 = 0.62 {nm}, p = E/c = (2.0×10³)(1.60×10^(-19))/(3.0×10^8) = 1.07×10^(-24) {kg,m,s^(-1)}.

(b) p = √(2mK) = √(2(9.11×10^(-31))(3.20×10^(-16))) = 2.41×10^(-23) {kg,m,s^(-1)} and λ = h/p = 2.75×10^(-11) {m} = 0.0275 {nm}. The check: K/mc² = 2 {keV}/511 {keV} = 0.004, so the non-relativistic formula is good to about 0.2%.

(c) For a photon p = E/c; for a massive particle p = √(2mE), which is much larger for the same energy because the mass is not zero. The electron's wavelength is therefore 23 times shorter, and it is the electron that can resolve atomic spacings — the reason electron microscopes reach sub-nanometre resolution while optical microscopes stop at about 200 {nm}. **Check:** 0.62/0.0275 = 22.5, close to √(2mc²/E) = √(2(511)/2) = 22.6.

Solution 10 — An electron in a box. Return →

(a) The walls force a node at each end, so L = nλ/2 and λ_n = 2L/n. Then p = h/λ = nh/(2L) and

$$E_n = (p^2)/(2m) = (n^2h^2)/(8mL^2). \quad (4)$$

(b) h²/(8mL²) = ((6.63×10^(-34))²)/(8(9.11×10^(-31))(1.0×10^(-9))²) = 6.03×10^(-20) {J} = 0.377 {eV}. So E_1 = 0.38 {eV} and E_2 = 4E_1 = 1.51 {eV}.

(c) Δ E = 3E_1 = 1.13 {eV}, so λ = 1240/1.13 = 1.1×10³ {nm} — the near infrared. **Check:** the zero-point energy E_1 is not zero, as the uncertainty principle demands, and the box confirms it: a smaller box raises E_1 as 1/L², so the electron in a 0.1 {nm} box would have E_1 of order 38 {eV}.